Supplementary Tables

Supplementary Table 1. Outcomes in Observational Studies[1]

Study	Number of Participants	Outcome	Scale	Description
Andreoli 2019	15	Swallowing function	Functional Oral Intake Scale[2]	(Mean (SD)): Pre = 3.07 (1.94), Post = 4.47 (2.26) Of those with aspiration and penetration 10/12 improved and of those with aspiration 7/10 resolved
		Feeding ability	Number of participants	For patients with gastrostomy dependence, improvement was noted in 7/8 patients; for patients without gastrostomy dependence, full oral feeds without restriction was achieved in 6/7 patients
Christiaanse 2011[36]	93	Swallowing function	Functional Oral Intake Scale[2]	Mean change (SD): Treatment = 0.77 (1.22), Control = 0.39 (0.98), however, this difference was not statistically significant
		Feeding ability	Number of participants	Treatment group: 80% of patients were fed by the same method before and after the intervention; 4 participants were no longer using enteral tubes at 6 month follow-up Control group: 70% of patients were fed by the same method before and after the intervention; 7 participants were no longer using oral tubes at 6 month follow-up
Ma 2019[32]	5	Swallowing function	Penetration Aspiration Scale[3]	Semisolid (Mean (SD)): Pre = 3.8 (1.5), Post = 2.1 (1.2) Liquid (Mean (SD)): Pre = 6.4 (2.1), Post = 4.3 (2.5)
Marcus 2019[33]	7	Swallowing function	Number of participants	All participants improved
		Child's feeding ability	Number of participants	5 of 7 participants improved Pre: 5 not safe for oral feeding (tube fed) Post: 3 full oral feeding, 2 partial oral feeding, 0 tube fed
		Impact on caregiver	Number of participants	5 of 7 caregivers felt feeding improved and were satisfied with the intervention; frequent need to travel to the hospital for treatments
Gao 2018[28]	62[4]	Child's feeding ability	TDS Classification	36 of 62 participants improved
			Efficacy	43 of 62 participants had an effective or significantly effective response to intervention
		Child's quality of life	Gessell	Adaptabilities: 7% improvement

		Swallowing function	Number of participants[5]	Language behaviour: 8% improvement Personal social development: 6% improvement Movement of lips and tongue: 6 of 62 participants improved Chewing movement: 2 of 62 participants improved Swallowing function: 5 of 62 participants improved Free eating: 7 of 62 participants improved
Rice 2012[34]	5	Swallowing function Child's quality of life	Number of participants Number of participants	All participants improved; Improved swallow safety and ability to advance in consistencies All participants improved

[1]Only studies without a control group were included in this table
[2]Higher scores indicate improvement
[3]Lower scores indicates improvement
[4]Number of participants included in our analysis (control group only)
[5]Number of participants in denominator is participants who exhibited this activity before treatment

Supplementary Table 2. Risk of Bias Assessment for the Included Observational Studies Using the Newcastle Ottawa Scale (NOS)[1]

Author	Year	Control Group	Selection (3/4 items)[2]	Comparability (1 item)	Outcome (3 items)	Score (Max = 7/8)[3]
Andreoli[35]	2019	No	★ ★ ★	★	★★	5
Ma[32]	2019	No	★			1
Marcus[33]	2019	No	★ ★ ★	★		4
Gao[28 4]	2018	No	★ ★	★	★	4
Rice[34]	2012	No	★ ★	★	★	4
Christiaanse[36]	2011	Yes	★ ★ ★ ★		★ ★	6

[1]A modified version of the Newcastle Ottawa scale was used (Appendix C).

[2]For prospective case series (with no control group), selection consists of 3 items. For retrospective studies (with control group), selection consists of 4 items.

[3]For prospective case series (with no control group), scores ranged from 0 (highest risk of bias) to 7 (lowest risk of bias). For retrospective studies (with control group), scores ranged from 0 (highest risk of bias) to 8 (lowest risk of bias).

[4]This study included NMES in both treatment and control groups, and therefore, was analyzed as a prospective case series.

Supplementary Table 3. Risk of Bias Assessment for the Included Randomized Controlled Trials Using the Cochrane Collaboration Tool[1]

		Selection bias		Performance bias		Detection bias	Attrition bias	Reporting bias	Other bias	
Author	**Year**	Random sequence generation	Allocation concealment	Blinding of participants	Blinding of personnel	Blinding of outcome assessment	Incomplete outcome data	Selective reporting	Similarity at baseline	Follow up
El-Sheikh[31]	2020	?	?	+	?	?	?	+	+	-
Lv[27]	2019	+	?	-	-	-	+	+	+	-
Serel Arslan[29]	2017	+	?	-	-	?	+	+	-	-
Song[30]	2015	?	?	+	-	-	+	+	+	-

[1] + indicates low risk of bias, - indicates high risk of bias, ? indicates unclear.